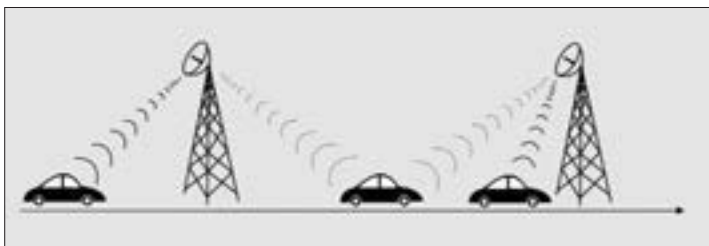


Every few seconds, your phone sends its MIN on the home control channel to tell the MTSO where it is.

When someone calls you, the request is first sent to the MTSO. The MTSO looks in its database to see which cell you're in. Once it has found you, it will decide on a frequency pair (one for talking and one for listening) for you to use for the call. It then tells your mobile phone via the control channel to tune itself to these frequencies. Once your phone has tuned itself, the MTSO connects the call.

On The Move

Being mobile, of course, means moving seamlessly from area to area while talking; and moving from area to area means moving from cell to cell in the network. Even when you're on a call, your mobile phone is telling your MTSO where it is. When you move towards the edge of a cell, the base station notes that your signal strength is weakening. At the same time, the base station for the cell you're approaching feels your presence growing. The base stations co-ordinate with each other through the MTSO, each telling it how close you are to which cell. Soon enough, the MTSO decides that it is time to switch to the new cell, and will select a new frequency pair for you to use. It then tells your phone to prepare to tune itself to this new frequency pair. Once your phone has switched to the new frequencies, you are connected to the new base station. This process is called a 'hand-off'. Sometimes the hand-off goes awry, and this is when your call gets inexplicably disconnected.



Moving between different coverage areas.